# Worked solutions: Q1·Q4

Manipulation Learning Lab | Instructor reference | 13 October 2026

**Run:** 20261013T051828Z_5a8ae81c, group_01. Dataset SHA256 prefix: 3f2eb92e0a50. The original four notebooks completed in 237.65 s; all 20 evaluations completed without runtime errors.

## Q1. What does the policy observe and command?

The 16 observations are end-effector world position (3), xyzw quaternion (4), two finger joint positions (2), and can world position plus xyzw quaternion (7). World Z points upward; positions and finger displacement are meters. The seven actions contain world translation increments (3), axis-angle rotation increments (3), and one gripper command (1). Normalized [-1,1] commands map to controller limits of ±0.05 m / ±0.5 rad; positive gripper commands close and negative open. Two observed finger joints still use one open/close action. Control runs at 20 Hz.

A trajectory shows approach, closure, lift and release order and provides observation/action training pairs. A final image cannot show this sequence. Human playback restores recorded states; it is not policy evaluation.

## Q2. How does behavior cloning learn?

A deterministic 256/256/256 multilayer perceptron predicts seven expert actions from each observation. Gradient updates minimize mean squared action error. Whole episodes are split 160/20/20; only training observations set normalization. Validation loss chooses the checkpoint, and test results never update weights or select the primary model. Only learning rate changes: seed 17, 50 epochs, batch 256, initialization, data, architecture and test cases are fixed. All four control checks passed. Both selected epoch 12; lr_a remained primary.

## Q3. Does lower training loss guarantee better manipulation?

| Variant | Final train MSE | Best val. MSE | Standard / shifted |
|---|---|---|---|
| lr_a: 0.001 | 0.017717 | 0.035376 | 3/5 / 3/5 |
| lr_b: 0.0003 | 0.020919 | 0.034123 | 5/5 / 5/5 |

No. lr_a has lower final training error but fewer stable placements. These rollouts use epoch-12 selected checkpoints, not epoch-50 weights. Closed-loop actions change future observations; prediction error on demonstration states does not measure recovery or task completion. Five cases and one training seed cannot establish a general learning-rate ranking.

## Q4. What changes under the position shift?

Each paired case keeps its source state and seed, adding 0.04 m to can world X. All success/failure labels stay the same here, but lr_a on demo_185 / seed 102 takes 141 standard versus 156 shifted steps (+0.75 s). A possible cause is altered approach geometry; timing alone cannot prove it. The coverage plot contains initial XY positions and paired-start lines, not trajectories or success labels. Position overlap can remain, and XY omits other inputs: this is not certified out-of-distribution testing.

# Worked solution: Q5 and evidence limits



## Q5. Locate a failure; distinguish fact, hypothesis and test

**Case:** primary student lr_a, standard_00, demo_39, seed 101; 500-step failure. **Stage:** grasp retention / beginning of lift. Contact is present at states 49–56 and first disappears at 57, remaining absent afterward. State 57 is the earliest directly supported contact-loss annotation, not a proven earliest root cause.

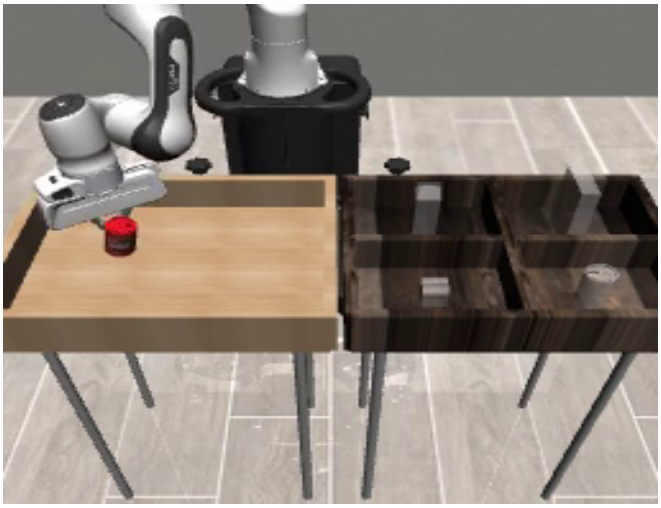
**Measured fact:** at state 57, opening is 0.022398 m, end-effector Z is 0.883759 m and can Z is 0.864667 m; the preceding action commands closure near +1. At 65, end-effector Z rises to 0.962711 m while can Z returns to 0.860271 m. Maximum lift is just 5.72 mm; the can never enters the target. The arm lifts without retaining it.

**Timing:** a[t−1] leads to s[t]. States 45, 53, 57 and 65 map to video frames 11, 13, 14 and 16, with zero offset. Videos record after every fourth action; state 0 has no exact video frame. Step 57 means 2.85 seconds of simulated control time.

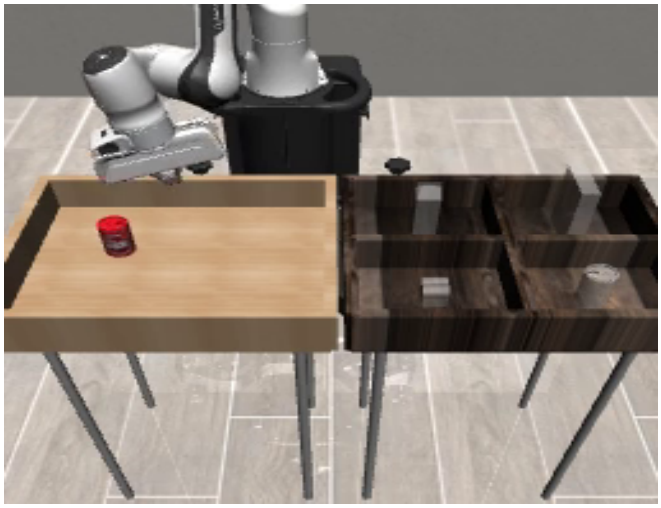
**Hypothesis:** approach alignment or closure/lift timing may prevent secure retention. Brief contact and a positive close command do not prove a stable grasp. The trace cannot distinguish these mechanisms or prove distribution shift.

**Proposed controlled test, not executed:** capture identical simulator/controller and random state just before closure. Compare the unchanged continuation against a predefined five-step closure hold at the pre-lift pose, then resume the same policy. Change only this timing intervention. Compare contact duration, relative can lift and stable placement on matched cases. Improvement would support a timing contribution; absence of improvement would leave other causes unresolved.

**Limits:** one fixed training seed, five paired cases per condition, simulation only. Stable placement requires target-margin containment, open gripper and no contact for a full second of continued policy execution, with displacement ≤1 cm and speed ≤5 cm/s. The recorded failure supports the fact of lost retention, not a guaranteed fix.



State 57 / frame 14: contact lost



State 65 / frame 16: empty lift

 Full logs, model hashes, videos and answers are included in the accompanying evidence.